

WEP - 02-05-2025

Topics: review of thermodynamics & gas dynamics

"Isentropic Flow with work"

$$\dot{q} = 0$$

$$\dot{w} \neq 0$$

$\dot{w} > 0$ for turbine

$\dot{w} < 0$ for compressor & pump

Note: The ideal gas law should not be used for pumps, as the working fluid is liquid.

An ideal pump is an adiabatic & reversible pump. $\rightarrow$ So ideal pump is isentropic

Ideal pump:

from thermo: $T ds = dh - \frac{dp}{\rho}$

$$dh_o = \frac{dp_o}{\rho}$$

$$\Delta h_o = \frac{\Delta p_o}{\rho}$$

1st law (Energy Eqⁿ)

$$h_{o2} - h_{o1} = \overset{0: \text{adiabatic}}{q} - w_p$$
$$= \frac{\Delta p_o}{\rho}$$

$$\Rightarrow w_p = -\frac{\Delta p_o}{\rho} \Rightarrow \text{"Pump work"}$$

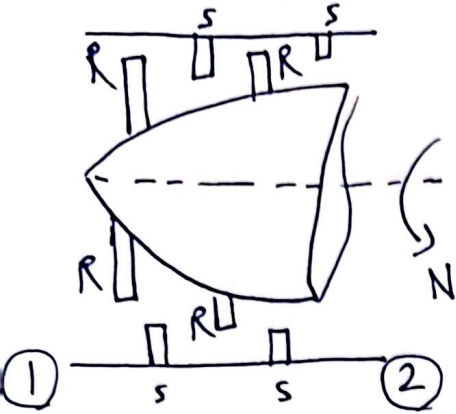
$$\dot{P}_{sp} = \dot{m} w_p = \dot{m} \left(-\frac{\Delta p_o}{\rho} \right) \quad \& \quad \Delta p_o = p_{o2} - p_{o1} \quad \& \quad p_{o2} > p_{o1}$$

$$\Rightarrow w_p \& \dot{P}_{sp} < 0$$

(P-1)

"Compressor"

Adiabatic & reversible compressor is called "ideal compressor"



R: rotor
S: stator

A compressor uses shaft power to generate a stagnation pressure increase

in the gas flowing through it.

$$\pi_c = \frac{P_{o2}}{P_{o1}} : \text{compression ratio } (P_o : \text{stagnation pressure})$$

$$\frac{T_{o2}}{T_{o1}} = \left(\frac{P_{o2}}{P_{o1}} \right)^{\frac{\gamma-1}{\gamma}} \quad (\text{isentropic relation})$$

$$\text{Energy Eq}^{\text{st}} : (\text{first law}) \quad h_{o2} - h_{o1} = \underbrace{q}_{\text{adiabatic}} - w_c$$

$$\Rightarrow w_c = -(h_{o2} - h_{o1}) = h_{o1} - h_{o2} = c_p (T_{o1} - T_{o2})$$

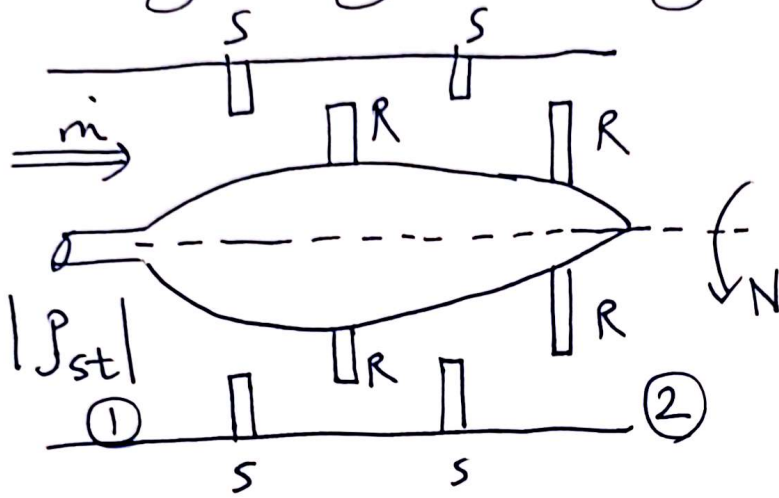
$$\dot{Q}_{sc} = \dot{m} w_c = \dot{m} c_p (T_{o1} - T_{o2})$$

$$\text{As } P_{o2} > P_{o1} \rightarrow T_{o2} > T_{o1} \rightarrow \begin{cases} w_c \\ \dot{Q}_{sc} \end{cases} < 0 \quad \left(\begin{array}{l} \text{receives} \\ \text{work} \end{array} \right)$$

(P-2)

"Turbine"

A turbine uses the stagnation pressure drop of the gas flowing through it to generate shaft power.



S: stator
R: rotor

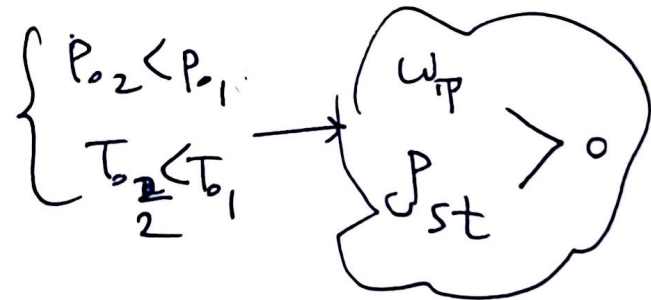
adiabatic & reversible turbine is called ideal turbine.

An ideal turbine is isentropic.

$\pi_t = \frac{P_{02}}{P_{01}}$ (turbine pressure ratio), $\frac{1}{\pi_t}$ is called turbine expansion ratio

Energy Eqⁿ: $h_{02} - h_{01} = q - w_t \Rightarrow w_t = -\Delta h_0 = c_p (T_{01} - T_{02})$
 q: adiabatic

$$\dot{P}_{st} = \dot{m} w_t = \dot{m} c_p (T_{01} - T_{02})$$



1.1), steady flow of perfect gas

conservation of mass: $\dot{m} = \dot{m}_1 = \dot{m}_2 = \dots$

Momentum Eqⁿ: $\dot{m}_2 u_2 = \dot{m}_1 u_1 + P_2 A_2 - P_1 A_1 + F$

(10.31)

F: body force & other surface force

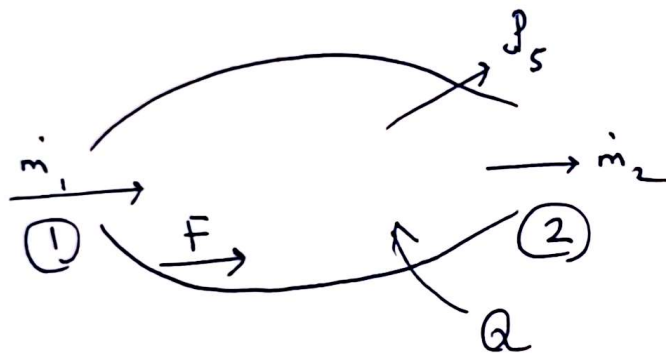
$$F = \underbrace{(\dot{m}_2 u_2 + P_2 A_2)}_{I_2} - \underbrace{(\dot{m}_1 u_1 + P_1 A_1)}_{I_1}$$

$$F = I_2 - I_1$$

I is impulse function

The force exerted on the device by the flow is of the same magnitude, but in opposite direction.

$$F_{\text{on device}} = I_1 - I_2 \quad \text{if } P = 0 \rightarrow I_1 = I_2$$



Energy Eqⁿ: (1st law)

$$\dot{m} \left(h_2 + \frac{u_2^2}{2} \right) = \dot{m} \left(h_1 + \frac{u_1^2}{2} \right) + \dot{Q} - \dot{P}_s$$

heat transfer
rate of work

Divide by $\dot{m}$:

$$\underbrace{\left(h_2 + \frac{u_2^2}{2} \right)}_{h_{o2}} = \underbrace{\left(h_1 + \frac{u_1^2}{2} \right)}_{h_{o1}} + \underbrace{\left(\frac{\dot{Q}}{\dot{m}} \right)}_q - \underbrace{\left(\frac{\dot{P}_s}{\dot{m}} \right)}_w$$

$$h_{o2} = h_{o1} + q - w$$

2nd law :

$$\dot{m} (s_2 - s_1) = \oint \dot{q} dA + S_{\text{gen}}$$

adiabatic flow: $\dot{q} = 0$

reversible flow: $S_{\text{gen}} = 0$

$s_1 = s_2$
isentropic flow

(P. 4)

2nd law: $s_2 - s_1 = c_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$

stagnation state: $s_{o2} - s_{o1} = c_p \ln \frac{T_{o2}}{T_{o1}} - R \ln \frac{P_{o2}}{P_{o1}}$

For isentropic flow $\rightarrow s_2 = s_1 \rightarrow \frac{P_2}{P_1} = \left(\frac{T_2}{T_1} \right)^{\frac{\gamma}{\gamma-1}}$

$s_{o2} = s_{o1} \rightarrow \frac{P_{o2}}{P_{o1}} = \left(\frac{T_{o2}}{T_{o1}} \right)^{\frac{\gamma}{\gamma-1}}$

@ $h_o = h + \frac{u^2}{2} \rightarrow c_p T_o = c_p T + \frac{u^2}{2} \rightarrow T_o = T + \frac{u^2}{2c_p}$

$u^2 = M^2 a^2 = M^2 \gamma R T$

$c_p = \frac{\gamma R}{\gamma-1}$

$\Rightarrow T_o = T \left(1 + \frac{\gamma-1}{2} M^2 \right)$

$\frac{T_o}{T} = 1 + \frac{(\gamma-1)}{2} M^2$

$\frac{P_o}{P} = \left(\frac{T_o}{T} \right)^{\frac{\gamma}{\gamma-1}} = \left(1 + \frac{\gamma-1}{2} M^2 \right)^{\frac{\gamma}{\gamma-1}}$